

Probability Theory and Statistics

Sample Midterm Solutions

1. **(Events A, B, C ; at least one always occurs; A and B independent; B and C disjoint.)** Given

$$\mathbb{P}(A \mid B) = \frac{1}{4}, \quad \mathbb{P}(C \mid A \cup C) = \frac{3}{4}, \quad \mathbb{P}(A \cap C) = \frac{1}{12}.$$

Solution. Independence gives $\mathbb{P}(A \mid B) = \mathbb{P}(A)$, hence $\mathbb{P}(A) = \frac{1}{4}$. From

$$\mathbb{P}(C \mid A \cup C) = \frac{\mathbb{P}((A \cup C) \cap C)}{\mathbb{P}(A \cup C)} = \frac{\mathbb{P}(C)}{\mathbb{P}(A \cup C)} = \frac{3}{4}$$

we get $\mathbb{P}(A \cup C) = \frac{4}{3} \mathbb{P}(C)$. Using Poincaré's formula,

$$\mathbb{P}(A \cup C) = \mathbb{P}(A) + \mathbb{P}(C) - \mathbb{P}(A \cap C) = \frac{1}{4} + \mathbb{P}(C) - \frac{1}{12},$$

thus $\frac{4}{3} \mathbb{P}(C) = \frac{1}{4} + \mathbb{P}(C) - \frac{1}{12}$, i.e. $\mathbb{P}(C) = \frac{1}{2}$. Since $A \cup B \cup C = \Omega$, we have

$$1 = \mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(A \cap B) - \mathbb{P}(B \cap C) - \mathbb{P}(C \cap A) + \mathbb{P}(A \cap B \cap C).$$

Here $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B) = \frac{1}{4}\mathbb{P}(B)$ (independence), and $\mathbb{P}(B \cap C) = \mathbb{P}(A \cap B \cap C) = 0$ (disjointness). Solving gives $\mathbb{P}(B) = \frac{4}{9}$.

2. **(Hungarian 32-card deck; Aladár draws 2, Bea draws 1 from Aladár's hand.)** Let B be the event “Bea's card is Red”, and for $i = 0, 1, 2$ let A_i be “Aladár has i Red cards”.

(a) *Find $\mathbb{P}(B)$.* By total probability:

$$\mathbb{P}(B) = \mathbb{P}(B \mid A_0)\mathbb{P}(A_0) + \mathbb{P}(B \mid A_1)\mathbb{P}(A_1) + \mathbb{P}(B \mid A_2)\mathbb{P}(A_2) = 0 \cdot \mathbb{P}(A_0) + \frac{1}{2}\mathbb{P}(A_1) + 1 \cdot \mathbb{P}(A_2).$$

With a 32-card deck (8 Red, 24 non-Red),

$$\mathbb{P}(A_1) = \frac{8 \cdot 24}{\binom{32}{2}} = \frac{12}{31}, \quad \mathbb{P}(A_2) = \frac{\binom{8}{2}}{\binom{32}{2}} = \frac{7}{124}.$$

$$\text{Hence } \mathbb{P}(B) = \frac{1}{2} \cdot \frac{12}{31} + \frac{7}{124} = \frac{1}{4}.$$

(b) *Find $\mathbb{P}(A_2 \mid B)$.* By Bayes:

$$\mathbb{P}(A_2 \mid B) = \frac{\mathbb{P}(B \mid A_2)\mathbb{P}(A_2)}{\mathbb{P}(B)} = \frac{1 \cdot \frac{7}{124}}{\frac{1}{4}} = \frac{7}{31}.$$

3. **(Two independent uniforms on $[0, 2]$ and $[0, 4]$; sum in $[3, 5]$.)** Geometric model on the rectangle $\Omega = [0, 2] \times [0, 4]$ (area $T_\Omega = 8$). The favorable region between the lines $x + y = 3$ and $x + y = 5$ within Ω has area $T_{\text{fav}} = \frac{7}{2}$ (by elementary geometry). Thus

$$\mathbb{P}(3 \leq X + Y \leq 5) = \frac{T_{\text{fav}}}{T_\Omega} = \frac{7/2}{8} = \frac{7}{16}.$$

4. **(Triple repetition over a BSC with crossover 1%; majority decoding.)** For one triple, the number of flipped bits is $B(3, 0.01)$. Majority is correct iff at most one error occurs, so

$$p_{\text{bit}} = \mathbb{P}(\text{bit decoded correctly}) = 0.99^3 + 3 \cdot 0.01 \cdot 0.99^2 = 0.999702.$$

Assuming independence across triples, the whole n -bit message is correct with probability $(p_{\text{bit}})^n$. We require $(0.999702)^n \geq 0.99$, equivalently

$$n \leq \frac{\ln 0.99}{\ln 0.999702} \approx 33.7209.$$

Hence the maximal integer n is $\boxed{33}$.

5. **(Density on $(1, c)$: $f(x) = \frac{(x-1)^2}{9}$; find c and $\mathbb{E}X$.)**

(a) $f \geq 0$. For a density we need $\int_{-\infty}^{\infty} f(x) dx = 1$, i.e.

$$1 = \int_1^c \frac{(x-1)^2}{9} dx = \frac{1}{9} \left[\frac{(x-1)^3}{3} \right]_1^c = \frac{(c-1)^3}{27}.$$

Thus $c-1 = 3$ and $c = \boxed{4}$.

(b) With $c = 4$,

$$\mathbb{E}X = \int_1^4 x \frac{(x-1)^2}{9} dx = \frac{1}{9} \int_0^3 (u+1)u^2 du = \frac{1}{9} \left[\frac{u^4}{4} + \frac{u^3}{3} \right]_0^3 = \boxed{\frac{13}{4}}.$$

- 6.* **($X \sim \text{U}[0, 2]$, $Y = \frac{1}{(1+X)^2}$.)**

(a) Let $X' = X + 1 \sim \text{Unif}[1, 3]$, with $F_{X'}(t) = \frac{t-1}{2}$ for $t \in [1, 3]$. For $y \in [1/9, 1]$,

$$F_Y(y) = \mathbb{P}\left(\frac{1}{(1+X)^2} < y\right) = \mathbb{P}\left(\frac{1}{(X')^2} < y\right) = \mathbb{P}\left(X' > \frac{1}{\sqrt{y}}\right) = 1 - F_{X'}\left(\frac{1}{\sqrt{y}}\right) = \frac{3}{2} - \frac{1}{2\sqrt{y}}.$$

Hence the density (by differentiation) is

$$f_Y(y) = \frac{1}{4y^{3/2}}, \quad y \in \left[\frac{1}{9}, 1\right],$$

and $f_Y(y) = 0$ otherwise. The expectation:

$$\mathbb{E}Y = \int_{1/9}^1 y \frac{1}{4y^{3/2}} dy = \int_{1/9}^1 \frac{1}{4y^{1/2}} dy = \frac{1}{4} \cdot 2 \left[y^{1/2} \right]_{1/9}^1 = \boxed{\frac{1}{3}}.$$

(b) Since F_Y is strictly increasing on its support, and

$$\frac{1}{(1+\mathbb{E}X)^2} = \frac{1}{(1+1)^2} = \frac{1}{4}, \quad \mathbb{E}Y = \frac{1}{3},$$

we have

$$\mathbb{P}\left(Y < \frac{1}{(1+\mathbb{E}X)^2}\right) = F_Y\left(\frac{1}{4}\right) = \frac{1}{2} \quad \text{and} \quad \mathbb{P}(Y < \mathbb{E}Y) = F_Y\left(\frac{1}{3}\right) = \frac{3}{2} - \frac{\sqrt{3}}{2} > \frac{1}{2}.$$

Thus $\boxed{\mathbb{P}(Y < \mathbb{E}Y) > \mathbb{P}(Y < (1+\mathbb{E}X)^{-2})}$.